

Department of Information Technology

PeopleAdmin Position Description Form

*Classification Title IT Security & Compliance Specialist I	*Working Title of Position Cyber Analyst Intern & Apprentice	*Beacon Position Number
Department/Agency NCDIT	Division ESRMO	Section/Unit
Clone of Existing Position?		If Yes, List Beacon Position Number
Position Justification / Reasons for Classification Action Request		
Key Changes in position since last classified:		
Present Classification Title / Salary Grade		Proposed Classification Title / Salary Grade
*Name of Employee		*Holder's Employee Number
*Name of Immediate Supervisor		*Supervisor's Position Title and Number
Work Schedule		Work Hours
*Primary Purpose of the Organizational Unit The Enterprise Security & Risk Management Office (ESRMO) is housed within the North Carolina Department of Information Technology (NCDIT) and is responsible for cybersecurity in the state. This office is led by the State Chief Information Security Officer who reports to the State Chief Information Officer. The ESRMO houses the state's cybersecurity and risk management subject matter experts who are responsible for providing and supporting a variety of functions within the state both directly to NCDIT as well as other state, county, and local entities and is organized to help the state meet the requirements of security standards legislation, N.C.G.S. § 147-33.110-33.113 and N.C.G.S. § 147-33.72c and other legal and regulatory requirements. The ESRMO provides several primary functions within the state which include: - Providing and managing a statewide information security governance and risk management program - Working closely with all NCDIT and agency executive leadership to identify and analyze emerging trends that may impact or present new risk to the state's information technology security so that tools, policies, procedures, plans, and strategies are constructed, implemented, adopted, and revised accordingly. - Governance in the development, delivery, and maintenance of a cybersecurity program, including the Statewide Information Security Manual that sets policy around safeguards for the state's information assets and supporting infrastructure against unauthorized use, disclosure, modification, damage or loss. This includes developing and managing a compliance program to ensure covered entities comply with current policies and procedures. - Coordinating with federal, state, and local entities, as well as critical infrastructure providers and the private sector, in North Carolina on a proactive cybersecurity posture that includes identifying and responding to cybersecurity risks and attacks. - Providing incident response support for state and other entities as necessary to protect state, county, and local entities and through membership and joint operation of the Joint Cybersecurity Task Force as authorized by Executive Order 254. - Identifying, evaluating, selecting, implementing, and managing cybersecurity services and tools that contribute to compliance with industry best-practices, state policies, and applicable law in effort to reduce cybersecurity risk. - Review of projects, contracts, and 3rd party risk related to information technology, and operational technology products and services being used within the state.		

- Providing training for risk management, business continuity, resiliency, cybersecurity, and other related subjects for employees of state agencies.
- Providing annual review and analysis of agency continuity plans.
- Conducting audit and other validation activities to determine levels of compliance and assist with remediation of any findings.
- Operational support through information security implementation, monitoring, threat and vulnerability management, analysis, mitigation processes & procedures, and cyber security incident management.
 - Help on cybersecurity matters to State, Local, Territorial, and Tribal (SLTT) government entities as required.
 - And other functions related to cybersecurity and resiliency as requested by the State Chief Information Officer and as defined through General Statute.

The ESRMO, as a part of NCDIT works closely with NCDIT staff and with executive branch agencies to ensure compliance with security statutes and mandates, statewide information security policies, the Statewide Technical Architecture, industry best practices, and other regulatory requirements. Working with state agencies, federal and local governments, citizens and private sector businesses, the ESRMO helps to manage risk in order to support secure and sustainable information technology services that meet the needs of the citizens of North Carolina.

***Primary Purpose of the Position**

The Cybersecurity Intern and/or Apprentice serves as an entry level technical expert within the Enterprise Security and Risk Management Office (ESRMO), responsible for supporting the Security Operations Center (SOC) and broader Cyber Threat Analysis Center (CTAC) which provides cybersecurity support beyond ESRMO to other state, local, tribal, and territorial (SLTT) entities. This role involves monitoring, analyzing, and responding to cybersecurity threats and incidents to enhance the overall cybersecurity posture of the state's information systems.

Reporting to the Deputy State Chief Information Security Officer, the Cybersecurity Intern and/or Apprentice works closely with other teams within ESRMO to support and augment their efforts in cybersecurity event review and analysis, vulnerability management, threat intelligence, incident response, and general cybersecurity operations. This position plays a critical role in ensuring the security and integrity of the state's information systems by conducting day-to-day operations of cybersecurity platforms and ensuring that tools are monitored, events are reviewed, analyzed, and results of those reviews are communicated to appropriate stakeholders.

This role has responsibility for a variety of cybersecurity analysis tasks, including threat detection, incident investigation, incident response support, vulnerability remediation support, and vulnerability assessment, to ensure that all security solutions work together effectively to protect the state's information systems. This role may also be called upon to provide these capabilities to other state, county, and local entities as directed. In addition, the Cybersecurity Intern and/or Apprentice is also responsible for collaborating with and supporting other teams within ESRMO and throughout the state to ensure proper operation of cybersecurity solutions in the respective environments.

***Knowledge, Skills, and Abilities Recommended in this Position**

- An understanding of cybersecurity products and services.
- Understanding of industry-standard security protocols and techniques, including but not limited to endpoint detection & response, vulnerability management, privilege account management, proxy and filtering, encryption technologies, multi-factor authentication, network firewalls, web application firewalls, intrusion detection/prevention systems, and various SIEM tools.
- Understanding of computer and network forensics, system and network security, incident management, intrusion detection, vulnerability and patch management, log analysis, and related technologies and how to effectively operationalize these tools and technology.
- Understanding of IT systems from operating systems, networking, and applications and how to apply cybersecurity to IT environments.

- Experience in designing and building reports, dashboards and other related capabilities in an enterprise.
- Knowledge of cybersecurity incident management, coordination, and resource management.
- Understanding how to handle sensitive and restricted information responsibly.
- Understanding of least privilege access, need to know, and two-person processes and concepts.
- The ability to develop and implement security policies, procedures, and standards.
- Problem-solving and process improvement skills.
- Ability to create and maintain documentation such as manuals, guides, frameworks, and runbooks.
- Demonstrated ability to work well on collaborative, cross-functional teams.

Desired

- Ability to stay current on emerging cybersecurity tools and technology such as the use of AI and Generative AI within cybersecurity and IT.
- Understanding of policy, regulation, and law applicable to cybersecurity.
- Understanding and use of attack and defense frameworks such as Mitre ATT&CK Framework, Diamond Model of Intrusion Analysis, and Cyber Kill Chain.
- Familiarity with ISO standard 27001, NIST 800-53, HIPPA, NIST Cybersecurity Framework, and other standards.
- Familiarity with Operational Technology (OT) environments, including ICS and SCADA systems.

***Education and Experience Required (E&E): Please use the official E&E on the class spec. Class specs can be found here: <https://oshr.nc.gov/state-employee-resources/classification-compensation/class-specs>**

2 full academic semesters of post high school study in Cybersecurity or a related IT field or closely related field from an appropriately accredited institution and one year experience in IT Security; or an equivalent combination of education and experience.

***License or Certification Required by Statute or Regulation**

***Description of Work**

Describe below in detail the major functions of this position as well as the duties and responsibilities required for each of those functions. Indicate the percentage of time the employee spends on each functional element. The percentage amounts should add up to 100 percent. In addition, indicate whether it is an essential duty/function as defined by the Americans with Disabilities Act.

The duties and responsibilities of a Cybersecurity Intern may include:

- Assisting in the implementation of security measures to protect computer systems, networks and data.
- Learning to anticipate security alerts, incidents and disasters and reduce their likelihood.
- Assisting in the analysis of software systems for vulnerabilities and risks.
- Monitoring system logs and identifying potential issues with computer systems.
- Assisting in the implementation of firewalls and data encryption programs to protect sensitive information.
- Participating in the development of security standards and best practices for the organization.
- Helping to educate fellow employees about computer security.
- Assisting in the preparation of reports that document security breaches and the extent of the damage caused by the breaches.
- Staying current on IT security trends and news.
- Performing routine assessments to ensure security measures are functioning correctly.

- Support the security operations center (SOC) and cybersecurity threat analysis center (CTAC) in reviewing, analyzing, and reporting on cybersecurity events and suggesting remediation options
- Assisting with the coordination of threat intelligence activities to identify and mitigate emerging threats.
- Assist with the development and implementation of security policies, procedures, and standards to ensure compliance with state and federal regulations.
- Work with other ESRMO leaders to ensure consistency and alignment of security policies across the organization.
- Conduct regular reviews and updates of security policies and procedures to address emerging threats and vulnerabilities.
- Consult with other agencies, SLTT entities, and partner entities on applicable N.C.G.S matters and assist with policy guidance and implementation as needed.
- Create reports as requested by leadership on the status of initiatives, effectiveness of operations, or fulfill requests for data in support of such reports.
- Work with the ESRMO team to fulfill audit and compliance requirements as needed.
- Assist state agencies with annual assessment requirements.
- Participate in the incident response team to assist with managing and mitigating security incidents within ESRMO as necessary.
- Assist with the coordination with federal, state, and local entities, as well as critical infrastructure providers and the private sector, to respond to security incidents as necessary.
- Assist with conducting post-incident reviews and develop recommendations for improving incident response processes.
- Participate in training, table-top exercises, and other relevant educational material.
- Collaboration with Security Liaisons to assist with identifying and implementing security tools and technologies and to ensure proper Security Operations support of represented entities.
- Assist with assessment, configuration, and implementation of security products and services as required to support the security operations team.

***= Required**

ADA Checklist

Checklist for physical activities and requirements, visual acuity, and working conditions of the position.

Physical Activity

☐	Climbing	Ascending or descending ladders, stairs, scaffolding, ramps, poles and the like, using feet and legs and/or hands and arms. Body agility is emphasized. This factor is important if the amount and kind of climbing required exceeds that required for ordinary locomotion.
☐	Balancing	Maintaining body equilibrium to prevent falling when walking, standing or crouching on narrow, slippery or erratically moving surfaces. This factor is important if the amount and kind of balancing exceeds that needed for ordinary locomotion and maintenance of body equilibrium.
☐	Stooping	Bending body downward and forward by bending spine at the waist. This factor is important if it occurs to a considerable degree and requires full use of the lower extremities and back muscles.
☐	Kneeling	Bending legs at knee to come to a rest on knee or knees.
☐	Crouching	Bending the body downward and forward by bending legs and spine.
☐	Crawling	Moving about on hands and knees or hands and feet.
☐	Reaching	Extending hand(s) and arm(s) in any direction.
☐	Standing	Particularly for sustained periods of time.
☐	Walking	Moving about on foot to accomplish tasks, particularly for long distances or moving from one work site to another.
☐	Pushing	Using upper extremities to press against something with steady force in order to thrust forward, downward, or outward.
☐	Pulling	Using upper extremities to exert force in order to drag, haul, or tug objects in a sustained motion.
☐	Lifting	Raising objects from a lower to a higher position or moving objects horizontally from position to position. This factor is important if it occurs to a considerable degree and requires the substantial use of the upper extremities and back muscles.
☐	Fingering	Picking, pinching, typing, writing or otherwise working, primarily with fingers rather than with the whole hand or arm as in handling.
☐	Grasping	Applying pressure to an object with the fingers and palm.
☐	Feeling	Perceiving attributes of objects, such as size, shape, temperature or texture by touching with skin, particularly that of the fingertips.
☐	Talking	Expressing or exchanging ideas by means of the spoken word. Those activities in which they must convey detailed or important spoken instructions to other workers accurately, loudly, or quickly.
☐	Hearing	Perceiving the nature of sounds at normal speaking levels with or without correction. Ability to receive detailed information through oral communication, and to make fine discriminations in sound.
☐	Repetitive Motions	Substantial movements (motions) of the wrists, hands, and/or fingers.
☐	Smelling	

Physical Requirements – CHECK ONLY ONE

☐	Sedentary Work	Exerting up to 10 pounds of force occasionally and/or a negligible amount of force frequently or constantly to lift, carry, push, pull or otherwise move objects, including the human body. Sedentary work involves sitting most of the time. Jobs are sedentary if walking and standing are required only occasionally and all other sedentary criteria are met.
☐	Light Work	Exerting up to 20 pounds of force occasionally and/or up to 20 pounds of force

		<i>frequently, and/or a negligible amount of force constantly to move objects. If the use of arm and/or leg controls requires exertion of forces greater than that for Sedentary Work and the worker sits most of the time, the job is rated for Light Work.</i>
☐	Medium Work	<i>Exerting up to 50 pounds of force occasionally, and/or up to 20 pounds of force frequently, and/or up to 10 pounds of force constantly to move objects.</i>
☐	Heavy Work	<i>Exerting up to 100 pounds of force occasionally, and/or up to 50 pounds of force frequently, and/or up to 10 pounds of force constantly to move objects.</i>
☐	Very Heavy Work	<i>Exerting in excess of 100 pounds of force occasionally, and/or in excess of 50 pounds of force frequently and/or in excess of 10 pounds of force constantly to move objects.</i>

Visual Acuity Requirements – CHECK ONLY ONE

☐	A	<i>The worker is required to have close visual acuity to perform an activity such as preparing and analyzing data and figures, transcribing; viewing a computer terminal; extensive reading; visual inspections involving small defects; small parts, and/or operation of machines (including inspection); using measurement devices; and/or assembly or fabrication of parts at distances close to the eyes.</i>
☐	B	<i>The worker is required to have visual acuity to perform an activity such as operated machines such as lathes, drill presses, power saws and mills where the seeing job is at or within arm's reach; performs mechanical or skilled trades of a non-repetitive nature such as carpenters, technicians, service people, plumbers, painters, mechanics, etc.</i>
☐	C	<i>The worker is required to have visual acuity to operate motor vehicles or heavy equipment.</i>
☐	D	<i>The worker is required to have visual acuity to determine the accuracy, neatness, and thoroughness of the work assigned (i.e., custodial, food services, general laborer, etc.) Or to make general observations of facilities or structures (i.e., security guard, inspection, etc.).</i>

Working Conditions

☐	The worker is subject to inside environmental conditions	<i>Protection from weather conditions but not necessarily from temperature changes.</i>
☐	The worker is subject to outside environmental conditions	<i>No effective protection from weather.</i>
☐	The worker is subject to both environmental conditions	<i>Both inside and outside as described above.</i>
☐	The worker is subject to extreme cold	<i>Temperatures typically below 12 degrees for periods of more than one hour. Consideration should be given to the effect of other environmental conditions such as wind and humidity.</i>
☐	The worker is subject to extreme heat	<i>Temperatures above 100 degrees for periods of more than one hour. Consideration should be given to the effect of other environmental conditions such as wind and humidity.</i>
☐	The worker is subject to noise	<i>There is sufficient noise to cause the worker to shout in order to be heard above the noise level.</i>
☐	The worker is subject to vibration	<i>Exposure to oscillating movements of the extremities or whole body.</i>
☐	The worker is subject to hazards	<i>Includes a variety of physical conditions, such as proximity to moving mechanical parts, moving vehicles, electrical current, working on scaffolding and high places, exposure to high heat or exposure to chemicals, extreme brightness, radiation, acidic substances, and allergens.</i>
☐	The worker is subject to atmospheric conditions	<i>One or more of the following conditions that affect the respiratory system of the skin: fumes, odors, dusts, mists, gases or poor ventilation.</i>

☐	The worker is subject to oils	<i>There is air and/or skin exposure to oils and other cutting fluids.</i>
☐	The worker is required to wear a respirator	
☐	The worker is frequently in close quarters, crawl spaces, shafts, man holes, small enclosed rooms, small sewage and water line pipes, and other areas which could cause claustrophobia	
☐	The worker is required to function in narrow aisles or passage ways	
☐	The worker is exposed to infectious diseases	
☐	The worker is required to function around prisoners or mental patients	
☐	None: The worker is not substantially exposed to adverse environmental conditions (such as in typical office or administrative work)	
☐	The worker is exposed to blood borne pathogens	<i>HIV/AIDS, Hepatitis B and Hepatitis C.</i>

Cognitive/Mental Capabilities

(Choose one in each category)

Comprehension	
☐	<i>Understand and apply routine verbal and/or written instructions</i>
☐	<i>Understand and apply non-routine verbal and/or written instructions</i>
☐	<i>Understand complex problems and collaborate to explore alternative solutions</i>
Organization	
☐	<i>Organize actions to complete sequential and/or routine tasks</i>
☐	<i>Organize and prioritize individual work schedule to manage multiple tasks and/or projects</i>
☐	<i>Organize and prioritize the work schedules of others to manage multiple tasks and/or projects</i>
Decision Making	
☐	<i>Make decisions that have an impact on the individual's work</i>
☐	<i>Make decisions that have an impact on the immediate work unit's operations and/or services</i>
☐	<i>Make decisions that have an impact on a department's credibility, operations, and/or services</i>
Communication	
☐	<i>Communicate and exchange routine/basic information</i>
☐	<i>Communicate and explain a variety of information</i>
☐	<i>Communicate in-depth information for the purpose of interpreting and/or negotiating</i>